

Exploratory Data Analysis (EDA) – Complete Practical Guide

This document provides a complete, dataset-agnostic checklist for performing Exploratory Data Analysis (EDA). It can be applied to ANY dataset before Machine Learning or business analysis.

1. Problem Understanding

- Define the problem statement clearly
- Identify whether it is classification, regression, or clustering
- Understand business or real-world objective

2. Data Loading & Initial Inspection

- Load dataset correctly
- Inspect first and last rows
- Understand column names and structure

3. Dataset Structure & Metadata

- Check number of rows and columns
- Inspect data types of each column
- Identify categorical vs numerical features

4. Target Variable Analysis

- Identify target variable
- Check class distribution
- Detect class imbalance

5. Data Quality Checks

- Check missing values
- Check duplicate records
- Detect invalid or inconsistent values

6. Data Cleaning

- Handle missing values appropriately
- Fix incorrect data types
- Remove or correct invalid records

7. Feature Categorization

- Separate numerical and categorical features
- Identify ordinal vs nominal categories

8. Univariate Analysis

- Analyze individual feature distributions

- Identify skewness and spread
- Detect outliers

9. Bivariate Analysis

- Analyze relationship between features and target
- Identify important features
- Understand dependency patterns

10. Multivariate Analysis

- Study interaction between multiple features
- Check correlation between variables
- Detect multicollinearity

11. Outlier Detection & Treatment

- Identify outliers using statistical methods
- Decide whether to keep, cap, or remove them

12. Feature Engineering

- Create new meaningful features
- Encode categorical variables
- Scale numerical variables

13. Data Transformation

- Apply normalization or standardization
- Log or power transformations if required

14. Insights & Observations

- Summarize key patterns discovered
- Write clear insights in plain language

15. Readiness for Machine Learning

- Ensure data is clean and consistent
- Ensure features are in numeric format
- Confirm no data leakage

Note: EDA is an iterative process. These steps may be revisited multiple times as new insights are discovered.